

PashuPoshan AI

Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers

PROBLEM STATEMENT DETAILS

Problem Statement ID: 26111

Ministry / Organization: Ministry of Fisheries, Animal Husbandry & Dairying (DAHD)

Theme: Agriculture, FoodTech & Rural Development

Target Users: Smallholder Dairy Farmers, Dairy Cooperatives (NDDB/AMUL), Livestock Extension Officers

TEAM & PROTOTYPE PROFILE

Team Name: [Insert Team Name]

Team Leader: [Insert Team Leader Name]

College / Institution: [Insert Institute Name]

Prototype Status: Fully Functional Offline-First PWA (Tested & Verified)

Core USPs: • Dual-Mode Optical & Strip Sensing • Flieg's Silage Scoring • BIS IS:2052 Grading • ICAR 4-Slot Precision Ration • 100% Offline PWA • 5 Languages

Problem Statement & Field Challenges

Understanding the critical bottlenecks faced by India's 80+ million dairy farmers under SIH PS 26111

1. RAMPANT ADULTERATION

Commercial cattle feed pellets are routinely adulterated with **synthetic urea (NPN)** to artificially inflate apparent Crude Protein tests, causing toxic rumen ammonia bloat. Cheap **sand/silica** (acid-insoluble ash > 5%) is blended as filler, causing severe abomasal impaction.

2. SILAGE FERMENTATION FAILURES

Over 40% of village bunker pits suffer from improper compaction and loose plastic seals. Infiltration of oxygen triggers aerobic heating (>38°C), clostridial rotting, and lethal **Aspergillus mold** growth, leading to direct **Aflatoxin M1 contamination** in milk.

3. SEVERE LABORATORY LAG & COST

Standard laboratory wet chemistry testing (Kjeldahl protein, Soxhlet fat) costs **₹1,200 – ₹2,500 per sample** and requires **10 to 14 days** turnaround. In rural cowsheds, farmers cannot pause daily feeding while waiting for district lab reports.

4. UNBALANCED DIETARY LOSSES

Testing feed in isolation does not solve cattle health. Farmers lack localized tools to formulate a composite daily **Total Mixed Ration (TMR)** tailored to indigenous (*Gir/Sahiwal*) vs. crossbred cattle, causing 15-20% daily milk yield losses.

Key Takeaway: India is the world's largest milk producer (230+ MT), yet smallholder dairy farmers lack any portable, zero-cost diagnostic screening tool at the cowshed gate. PashuPoshan AI solves this gap directly.

Proposed Solution: PashuPoshan AI

An offline-first, dual-sensing mobile platform providing rapid screening, safety triage, and precision ration balancing

1. DUAL-MODE RAPID SENSING

- **Camera Vision:** Client-side HTML5 Canvas pixel extraction for surface texture variance and dark mold clustering.
- **Colorimetric Strips:** Rapid chemical test-strip analysis using Euclidean color distance (ΔE) in color space to detect synthetic urea and silage pH.

4. ANIMAL SAFETY & EMERGENCY TRIAGE

Eliminates dangerous DIY medical advice. Provides immediate batch isolation protocols and a **1-tap call button to the National Animal Disease Emergency Helpline (1962)** for suspect feeds.

2. FLIEG'S SILAGE QUALITY INDEX

- Implements the scientific **Flieg Scoring Model (0–100)** combining pH and Dry Matter (DM%):
- **Score 81–100:** Premium lactic fermentation.
 - **Score < 40:** Spoiled / High Butyric acid clostridial risk requiring immediate feed suspension.

5. TRUE OFFLINE-FIRST PWA ENGINE

Works 100% offline inside rural cowsheds. Stores compressed images in **IndexedDB** (avoiding LocalStorage quota crashes), with Service Worker v2 asset caching and a transparent offline sync queue.

3. CLOSED-LOOP ICAR RATION BALANCER

Doesn't stop at testing. The tested sample dynamically populates its specific nutritional slot in an **ICAR (2013) 4-Slot Total Mixed Ration** (Green Fodder + Dry Straw + Concentrate + Minerals) tailored to breed and daily milk yield.

6. VERNACULAR & ACCESSIBLE

100% translated into **Hindi, Marathi, Gujarati, Punjabi, and English**. Features text-to-speech audio playback for low-literacy farmers and clean printable A4 PDF field screening reports.

System Architecture & End-to-End Workflow

Modular, client-side edge computing pipeline designed for zero-latency, offline field execution

• Mobile Camera Viewfinder • Gallery photo upload • Chemical test-strip dip • Silage bunker parameters • Animal profile & milk yield	• Offscreen HTML5 Canvas • RGB & Luminance extraction • Dark mold clustering (<60) • ΔE Color distance matching • Image compression (JPEG 0.6)	• BIS IS:2052 Table 1 check • Flieg Score (pH + DM%) • Urea spike detection (>0.5%) • Acid insoluble ash limit check • Aflatoxin M1 mold screening	• ICAR 4-Slot TMR balancer • Live DMI / CP / TDN math • 1962 Veterinary Helpline • Text-to-Speech audio guide • Printable A4 PDF certificate
Storage: Camera / File API	Engine: Canvas 2D Context	Standards: BIS / ICAR / FSSAI	Persistence: IndexedDB + PWA

Edge Computing Advantage: All canvas image analysis and nutritional calculations run 100% on the mobile device client-side. Zero cloud latency, zero server bandwidth cost per test, and zero dependency on rural cellular connectivity.

Scientific Rigor & Regulatory Alignment

Direct compliance with statutory Indian dairy standards and international peer-reviewed literature

STANDARD / REFERENCE	STATUTORY CLAUSE / EQUATION	HOW PASHUPOSHAN AI IMPLEMENTS IT
BIS IS:2052:2009 (Reaffirmed 2020) Compounded Feeds for Cattle	Table 1 Specifications: - Type II Concentrate: Min 20.0% Crude Protein (CP) - Moisture: Max 11.0% - Acid Insoluble Ash: Max 3.5%	Strict Compliance Engine: A 19.8% CP sample strictly fails Type II compliance and is graded <i>Tier B: Sub-Standard</i> (verified by regression test). Does not round up.
Flieg's Index & Kung et al. (2018) Silage Fermentation	Score = 220 + (2 × %DM - 15) - 40 × pH - Optimal Maize pH: 3.8 – 4.2 - Clostridial Spoilage threshold: pH > 4.8	Mathematical Scoring: Evaluates silage fermentation into 5 discrete tiers (Excellent, Good, Fair, Poor, Very Bad) and triggers aerobic heating alerts at >38°C.
ICAR (2013) 3rd Ed. Nutrient Requirements of Cattle & Buffalo	Dry Matter Intake (DMI): - Indigenous (*Gir/Sahiwal*): 2.5% Body Weight - Crossbred (*HF/Jersey*): 3.0% Body Weight - Maintenance: 0.8g CP / kg BW + 45g CP / kg milk	Composite 4-Slot TMR Engine: Dynamically sums DMI, CP (g), and TDN energy across Green Fodder, Dry Straw, Concentrate, and Mineral Mixture per breed.
FSSAI Regulations (2011) Contaminants, Toxins & Residues	Aflatoxin M1 Ceiling: Maximum permissible limit of 0.5 µg/kg in fluid milk (transmitted from mouldy forage).	Hazard Triage: Dark mold clustering in silage flags Tier C hazard; instructs immediate feed segregation to prevent Aflatoxin M1 milk contamination.

Honest Engineering, Safety & Offline Hardening

Prioritizing animal health safety, transparent heuristics, and rugged cowshed dependability

1. CLINICAL SAFETY OVER DANGEROUS DIY

- Competitor prototypes often give dangerous medical advice (e.g. prescribing acetic acid drenches for urea bloat).
- **Our Protocol:** Immediate feed withdrawal, segregation of suspect bags, and 1-tap call to the **National Animal Disease Emergency Helpline (1962)**.

3. INDEXEDDB IMAGE STORAGE (NO CRASHES)

- Storing camera photos in browser `localStorage` causes `QuotaExceededError` crashes after 3 photos.
- **Our Engine:** Automatically compresses photos (max 400x300 JPEG 0.65) and stores blobs in **IndexedDB** (`pashuposhan\_images\_db`).

2. TRANSPARENT HEURISTIC CLAIMS

- No fake claims of '99% lab accuracy' on raw camera pixels.
- Every live camera scan is permanently badged: **'Prototype Heuristic — No Laboratory Measurement'**.
- Separates live camera estimates from certified NABL chemical wet testing.

4. HONEST OFFLINE SYNC QUEUE

- The app does not pretend to sync with a non-existent government cloud.
- Queued offline records remain preserved locally with a visible queue inspector, allowing manual verification or clearing.

Ethical AI Commitment: PashuPoshan AI adheres strictly to the Precautionary Principle in Animal Health. Screening tools must empower farmers to detect bad batches early without replacing registered veterinary professionals.

Competitive Landscape & Innovation Edge

How PashuPoshan AI compares against conventional laboratory testing and emerging portable spectrometers

EVALUATION CRITERIA	DISTRICT WET LABS (NABL)	PORTABLE NIR SPECTROMETERS	GENERIC DAIRY APPS	PASHUPOSHAN AI (OURS)
Turnaround Time	10 – 14 Days	5 – 10 Minutes	N/A (No testing)	Instant (< 3 Seconds)
Cost per Sample	■1,200 – ■2,500	■2.5 Lakhs hardware cost	Free / Subscription	■0 (Zero Hardware Cost)
Chemical Adulteration	Yes (Wet Chemistry)	Limited (Spectra library)	No	Yes (Colorimetric Strips)
Silage Flieg Scoring	Requires separate acid test	Complex calibration	No	Automated (pH + DM%)
Closed-Loop Ration Balancer	No (Disconnected PDF)	No (Raw data only)	Manual entry only	Dynamic (Tested Sample in TMR)
Rural Cowshed Offline	No	Requires Bluetooth bridge	Requires Internet	100% Offline PWA + IndexedDB

Impact, Farmer ROI & Cooperative Value

Measurable economic returns for dairy farmers and quality assurance for cooperative milk unions

FARMER LEVEL (HOUSEHOLD LEVEL)

- **+10% to +15% Milk Yield:** Balanced TMR ensures optimum microbial protein synthesis in the rumen without feed wastage.
- **■2.20 Savings per Liter:** Replacing overfed expensive concentrates with high-grade home-grown silage reduces daily feeding cost.
- **Elimination of Toxic Bloat:** Early detection of spiked urea saves **■35,000+** per milch cow in veterinary emergency treatments.

DAIRY COOPERATIVE LEVEL (NDDB / AMUL)

- **Clean Milk Compliance:** Prevents Aflatoxin M1 transmission into cooperative milk silos, protecting export quality.
- **Feed Supply Chain Audit:** Traceability of compound feed batches via QR code verification stops sub-standard suppliers.
- **Greenhouse Gas Mitigation:** ICAR precision balancing reduces enteric methane emissions per kg of milk produced by 12-15%.

+1.5 to 2.0 L
Daily milk yield boost per cow

< 3 Seconds
On-farm screening turnaround

100% Offline
Cowshed PWA operation

■0 Hardware
Zero capital barrier for farmers

Technology Stack & Engineering Rigor

Built with enterprise-grade TypeScript, resilient offline storage, and automated regression testing

FRONTEND & SENSING	OFFLINE STORAGE & PWA	AUTOMATED QUALITY ASSURANCE
• React 18 & TypeScript: Strict typing with zero <code>`any`</code> casts. • Tailwind CSS: Modern, responsive mobile shell with WCAG 2.1 touch sizing ($\geq 44\text{px}$). • HTML5 Canvas API: Client-side pixel parsing and colorimetric distance calculation.	• Service Worker v2: Cache-First runtime caching for all Vite JS/CSS chunks. • IndexedDB: High-capacity compressed image storage (<code>`pashuposhan_images_db`</code>). • LocalStorage Fallback: Quota-protected persistence for herds and logs.	• Vitest Suite: 10 automated unit tests covering: ✓ Flieg score math & pH clamping ✓ BIS IS:2052 19.8% CP failure regression ✓ ICAR DMI breed equations ✓ Prototype disclaimer validation • Build: Clean Vite compilation in 23s.

AUTOMATED TEST STATUS: 10 Passed (100%) across 3 test suites | 0 Errors, 0 Warnings | Error Boundary Runtime Crash Protection Active

Roadmap, Hardware Scaling & Conclusion

Phased deployment roadmap aligning with the National Livestock Mission and Rashtriya Gokul Mission

PHASE 1: CURRENT PROTOTYPE (COMPLETED) • Standalone PWA with camera vision & test strip analysis. • Flieg's score & BIS compliance grading engine. • ICAR 4-slot composite precision ration balancer. • 100% offline IndexedDB persistence & 5 vernacular languages. • Automated unit test suite & Printable A4 screening report.	PHASE 2: COOPERATIVE INTEGRATION (3–6 MONTHS) • Cloud sync API with Primary Dairy Cooperative Societies (DCS). • Computer vision fine-tuning on ground-truth NIR feed datasets. • Geo-tagged community adulteration outbreak heatmaps. • Integration with INAPH (National Animal Productivity & Health).	PHASE 3: IOT HARDWARE PROBE (6–12 MONTHS) • Ultra-low-cost BLE stainless-steel probe (■800). • Penetrates 1-meter into bunker pits to measure real-time core temperature, pH, and moisture. • Automatic Bluetooth synchronization with the PashuPoshan mobile app.
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Conclusion: PashuPoshan AI democratizes laboratory-grade dairy feed science for every Indian farmer. By uniting rapid sensing with ICAR precision nutrition, we protect animal health, ensure clean milk, and empower rural livelihoods.